MACHINE LEARNING ENGINEER ASSESSMENT

Freight Rate Prediction Challenge

A leakage-aware, production-style solution for 12,000 future loads

$123.25	5.68%	0.820	51.4%
OCTOBER MAE	OCTOBER MAPE	R-SQUARED	MAE IMPROVEMENT

Candidate Repository

eraj22

github.com/eraj22/freight-rate-ml-assessment

Development data

48,000 loads | Jan-Oct 2025

Final scoring data

12,000 loads | Nov-Dec 2025

Submission status: 12,000 prediction IDs validated, 31 fixed December rows validated, unit tests passing, and the employer-provided chart generated.

Prepared as a reproducible employer-facing assessment with code, diagnostics, documentation, CI, and final deliverables.

01

Executive summary

The decision, validation design, and headline result

The final solution is a three-seed CatBoost ensemble trained on log1p(posted_rate). The design is optimized for forward-looking accuracy, unseen route combinations, missing data, nonlinear distance effects, and rare extreme rates.

$123.25	$39.73	5.68%	0.820
HOLDOUT MAE	MEDIAN ABSOLUTE ERROR	HOLDOUT MAPE	HOLDOUT R-SQUARED

What matters

Temporal integrity

September selected the approach. October remained the untouched forward holdout.

Generalization

Geography and physical features support eight unseen cities and 736 unseen lanes.

Robust target

The log target retains every rate while limiting the influence of rare extremes.

Validation timeline

Stage	Training window	Evaluation window	Purpose
Development	2025-01-01 to 2025-08-31	September 2025	Select model and features
Final holdout	2025-01-01 to 2025-09-30	October 2025	Simulate a future month
Final fit	2025-01-01 to 2025-10-31	Nov-Dec 2025	Generate submission predictions

Why not random cross-validation? Random folds would mix future observations into past training windows. The chronological design better estimates the real deployment decision.

02

Data audit

Quality findings, anomaly treatment, and generalization risk

48,000	12,000	0	8
LABELED ROWS	FINAL ROWS	DUPLICATE IDS	UNSEEN CITIES

Quality review

Check	Labeled	Final	Treatment
Duplicate rows	0	0	No action
Duplicate load_id	0	0	Hard validation failure
Missing weight	300	165	Missing flag and native null handling
Negative weight	292	145	Absolute value plus anomaly flag
Missing market index	374	249	Missing flag and native null handling
Unseen lanes	-	736	Use physical and geographic features

Target behavior

Posted rate is right-skewed: median $2,030.76, mean $2,373.98, 99th percentile $5,972.83, and maximum $25,533. Extreme values are retained because the brief does not identify them as bad records.

Generalization controls

Unseen routes

Lane identity is supplementary. Distance, coordinates, equipment, weight, and market signals remain available.

No target leakage

Feature construction never uses posted_rate. Every transformation is row-local or training-history based.

Future scenario

December coordinates come from labeled cities; unavailable market signals use recent training medians.

Source: provided train_test.csv, validation.csv, and assessment brief. Counts are reproduced by scripts/data_audit.py.

03 Model and feature engineering

A robust tabular ensemble designed for future-month inference

01	02	03	04
AUDIT	**ENGINEER**	**ENSEMBLE**	**VALIDATE**
Validate IDs, missingness, ranges, and chronology.	Build route, calendar, quality, and interaction features.	Average three CatBoost models on the log target.	Freeze the pipeline and score the future October holdout.

Why CatBoost

Mixed data

Processes route categories alongside continuous load, geography, and market variables.

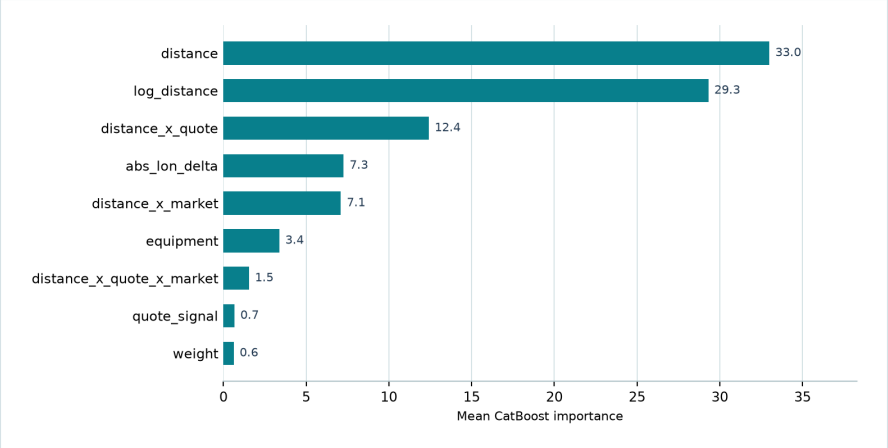
Operational robustness

Handles missing numeric values and unseen category combinations without a fragile one-hot pipeline.

Feature groups

Group	Examples	Purpose
Route	pickup, delivery, lane, coordinates	Geographic and directional effects
Load	distance, equipment, cleaned weight	Physical movement characteristics
Market	market index, quote signal	Observed pricing conditions
Interactions	distance x quote, distance x market	Convert per-mile signals to load scale
Calendar	month, weekday, cyclic terms	Seasonal and weekly patterns
Quality	missing and negative-value flags	Preserve anomaly information

Top feature importance



Importance is averaged across the final three models. It is descriptive rather than causal.

04

Final holdout results

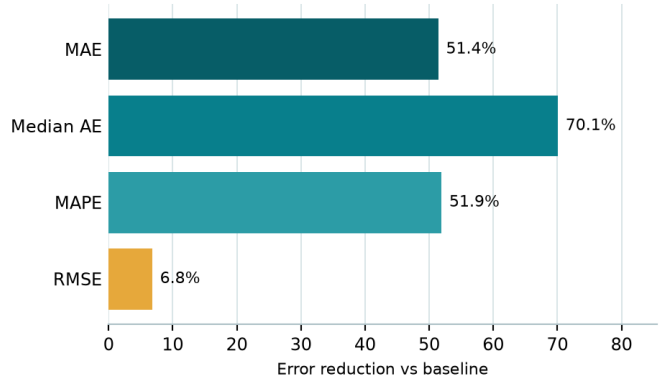
Performance on 4,853 unseen October loads

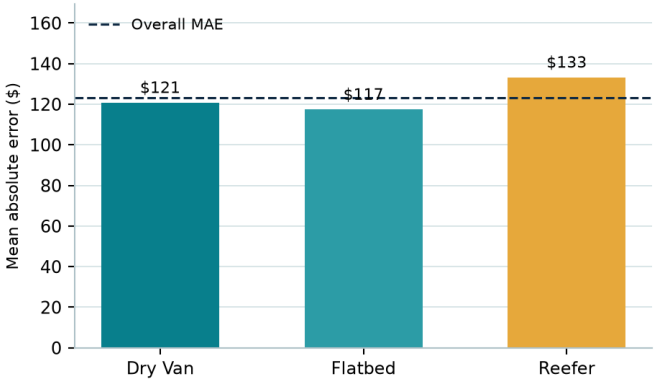
The final holdout was not used to fit the evaluated models. The baseline multiplies each distance by the median historical rate per mile from the training window.

$123.25
ENSEMBLE MAE

$253.85
BASELINE MAE

51.4%
MAE REDUCTION





Metric	Ensemble	Baseline	Change
MAE	$123.25	$253.85	-51.4%
RMSE	$647.62	$695.05	-6.8%
MAPE	5.68%	11.81%	-6.14 pp
Median AE	$39.73	$132.82	-70.1%
R-squared	0.820	0.793	+0.027

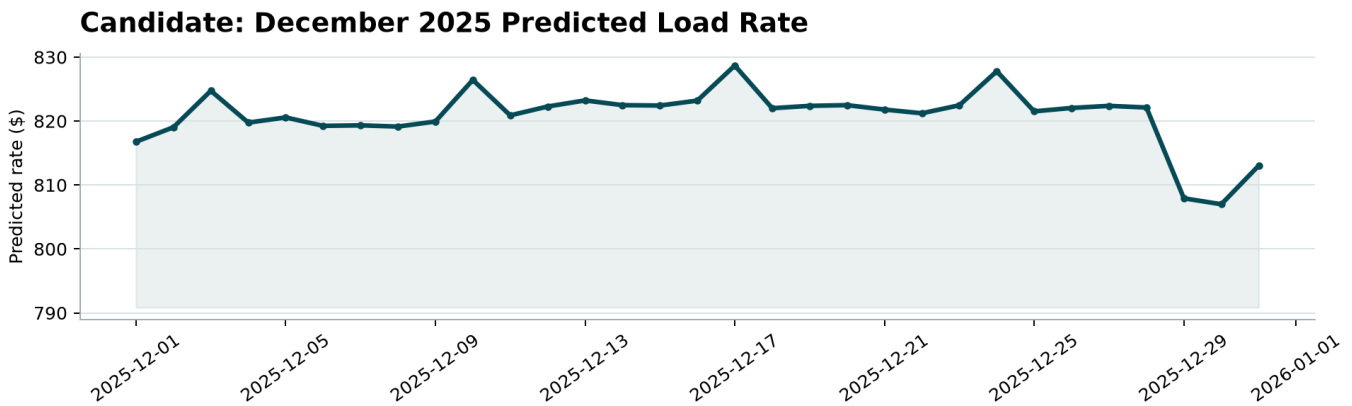
Typical-load accuracy is strong: median absolute error is $39.73, and 90% of absolute errors are below $166.64. RMSE remains higher because a small number of extreme rates create large squared errors.

The scorer successfully validates all 12,000 prediction IDs and all 31 fixed December scenario rows.

05

Fixed December prediction

The employer-provided scorer output and scenario assumptions



Fixed inputs: Lexington to Fort Wayne | 360 miles | Dry Van | 32,000 lb | only date changes

Scenario inputs

Lexington	Fort Wayne	360 mi	Dry Van	32,000 lb
PICKUP	DELIVERY	DISTANCE	EQUIPMENT	WEIGHT

Operational assumptions

Coordinates

Lexington and Fort Wayne use median city coordinates from the labeled data.

Market signals

Unavailable future market and quote signals use the latest 30-day training medians.

Date-only change

All physical inputs remain fixed, isolating learned calendar variation across 31 days.

The completed scenario contains **31 daily predictions**, ranging from **$806.99** to **$828.69**. The supplied score.py generated the chart after validating every row.

06

Production readiness

Reproducibility, controls, monitoring, and submission checklist

Repository controls

Deterministic

Fixed seeds, frozen hyperparameters, and exact dependency ranges.

Validated

Schema checks, positive-rate checks, ID alignment, unit tests, and CI.

Documented

README, model card, data audit, diagnostics, report, and Loom script.

Run sequence

Step	Command	Output
1	python -m pip install -r requirements.txt	Environment
2	python scripts/data_audit.py	Data-quality audit
3	python evaluate.py	Holdout metrics and diagnostics
4	python train_predict.py	Final prediction CSVs
5	python score.py --predictions validation_predictions.csv --december-predictions data/december_chart_inputs.csv	Validated chart
6	python -m unittest discover -s tests -v	Passing unit tests

Monitoring and limitations

Monitor MAE, MAPE, and error by equipment and lane after actual rates arrive. Retrain when the freight market or feature distributions drift. Keep human review for unusually expensive loads and predictions outside the observed operating range.

Submission checklist

Deliverable	Status	Location
Accessible GitHub repository	Ready	github.com/eraj22/freight-rate-ml-assessment
12,000 final predictions	Validated	validation_predictions.csv
Assessment report	Complete	reports/freight_rate_assessment_report.pdf
Fixed December chart	Generated	scorer_results/candidate_december.png
Camera-on Loom	Candidate records	Use reports/loom_walkthrough_script.md

Final artifacts are reproducible from the public repository using Python 3.12.